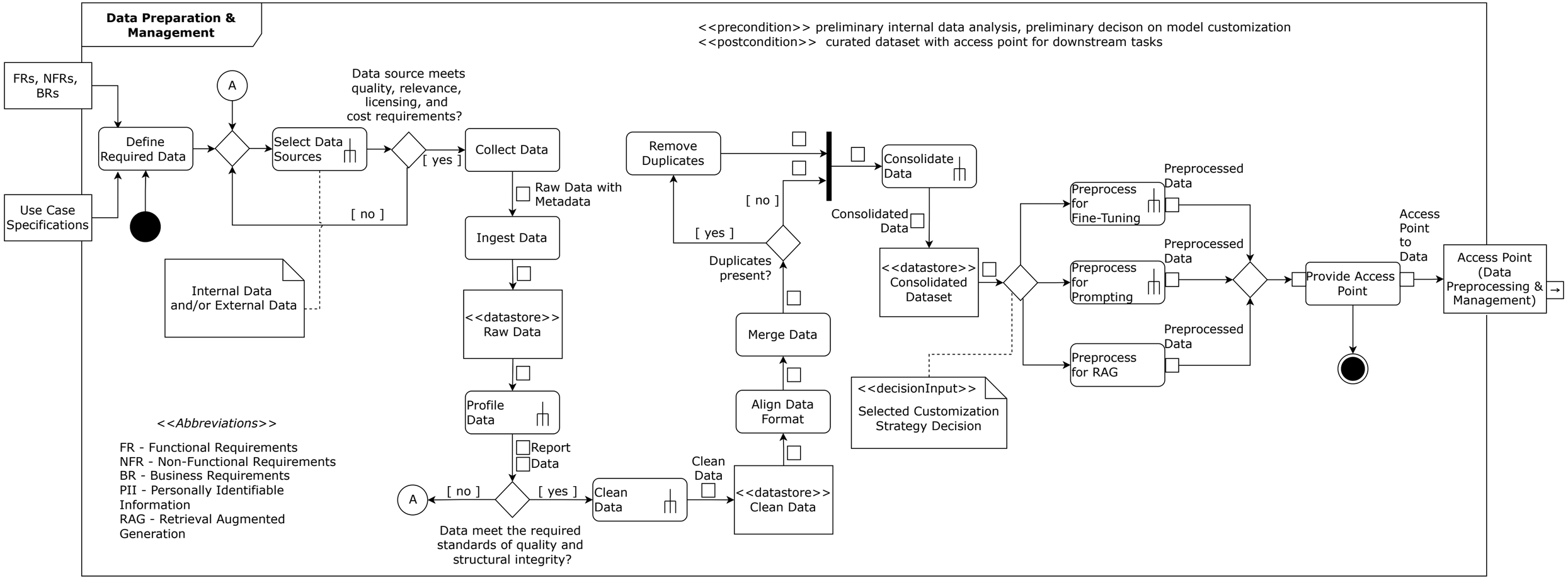
Data Preparation & Management
<<precondition>> preliminary internal data analysis, preliminary decison on model customization
<<postcondition>> curated dataset with access point for downstream tasks
FRs, NFRs, BRs
Use Case Specifications
Define Required Data
A
Select Data Sources
Data source meets quality, relevance, licensing, and cost requirements?
[yes]
[no]
Collect Data
Raw Data with Metadata
Ingest Data
<<datastore>> Raw Data
Internal Data and/or External Data
Profile Data
Report Data
[no]
[yes]
A
Data meet the required standards of quality and structural integrity?
Clean Data
Clean Data
<<datastore>> Clean Data
Align Data Format
Merge Data
Duplicates present?
[yes]
[no]
Remove Duplicates
Consolidate Data
Consolidated Data
<<datastore>> Consolidated Dataset
<<decisionInput>>
Selected Customization Strategy Decision
Preprocess for Fine-Tuning
Preprocess for Prompting
Preprocess for RAG
Preprocessed Data
Preprocessed Data
Preprocessed Data
Provide Access Point
Access Point to Data
Access Point (Data Preprocessing & Management)
<<Abbreviations>>
FR - Functional Requirements
NFR - Non-Functional Requirements
BR - Business Requirements
PII - Personally Identifiable Information
RAG - Retrieval Augmented Generation